



BOHIO SYSTEMS × KINAN

Setup guide

Running the Bohio agent with an EstateMaster trial, Outlook, Claude and OpenAI

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This guide does not rely on the EstateMaster manual, which could not be read from the build environment. Every step that depends on how EstateMaster behaves is marked "confirm in your trial": check it in the software, and with Altus where needed.

What you will have at the end

PHASE	WHAT IT GIVES YOU	TIME
1. Deploy the demo on Vercel	The agent online with Claude and/or OpenAI, keys kept on the server	30 min
2. Prepare the EstateMaster trial	A model whose inputs and outputs are linked to an Excel control workbook, and a mapped assumption register	half a day
3. Run the EstateMaster runner	A service on the EstateMaster machine that writes inputs, recalculates and reads outputs on request	half a day
4. Connect Outlook	Real emails read four times a day; assumption changes proposed for approval	1–2 hours (needs a Microsoft 365 admin)
5. Optional: WhatsApp, Excel/Copilot, scheduling	Channels and unattended scans	depends

Know what stays simulated. After phases 1 and 4 the agent reads real email and uses real AI, but its numbers still come from the replica in the page. Phase 3 gives you a working runner on the EstateMaster machine; pointing the agent's calculations at it is the next development step (a small change in the page, listed at the end). Use phase 2–3 to prove the round trip on your trial and time it.

Before you start

ITEM	NOTES
Windows PC or VM with the EstateMaster trial installed	EstateMaster is Windows desktop software. Keep this machine on while testing.
Microsoft Excel (desktop) on the same machine	Needed for the live-linked control workbook and the runner.
Python 3.11+ on the same machine	For the runner starter kit.
Vercel account	Free tier is enough to start; scheduled jobs may need a paid plan (check your plan's cron limits).
Anthropic API key and/or OpenAI API key	From the providers' consoles. One is enough; both give failover.
Microsoft 365 tenant with a Global or Application Administrator	For the Outlook connection (app registration and admin consent).
The zip <code>bohio-estatemaster-vercel.zip</code>	Contains <code>index.html</code> , <code>api/llm.js</code> , <code>api/scan.js</code> , <code>runner/</code> , <code>vercel.json</code> , <code>README.md</code> .

Phase 1: Deploy the demo on Vercel

- Unzip `bohio-estatemaster-vercel.zip`. You get a folder `bohio-estatemaster-demo`.
- Install the Vercel command line (needs Node.js): `npm i -g vercel`. In the folder run `vercel`, log in, accept the defaults (framework: Other, no build command). Alternatively push the folder to a GitHub repository and import it at `vercel.com/new`.
- In the Vercel project open **Settings** → **Environment Variables** and add the variables below, then redeploy (`vercel -prod`) so they take effect.
- Open the site. The Router button (top right) should read "Router · AI: Claude" (or OpenAI). Ask "should we approve this deal?" in the agent chat: the answer carries a red AI badge.
- If you set an access code, open Router, enter it under Access code, and save.

VARIABLE	VALUE	NEEDED FOR
ANTHROPIC_API_KEY	your Anthropic key	Claude
OPENAI_API_KEY	your OpenAI key	OpenAI
DEMO_PASSWORD	any access code	Recommended: stops strangers spending your credits
ANTHROPIC_URL , OPENAI_URL	optional gateway URLs	Corporate gateways only

Check the OpenAI model names under Router → Routing rules and models (e.g. `smart=anthropic:claude-sonnet-5-5`, `openai:gpt-5`) match models your account can use.

Phase 2: Prepare the EstateMaster trial

Goal: one project in EstateMaster whose key inputs can be written from Excel and whose outputs can be read in Excel, so software can drive it. Altus describes an Excel live-link that sends data between EstateMaster models and external workbooks; how it is set up and refreshed in your version is what this phase establishes.

2.1 A model to work with

1. Open the trial and create or open a feasibility project. A copy of a real KINAN model is ideal; otherwise build a simple mixed-use scheme similar to Al Narjis (residential for sale plus retail for hold).
2. Save a copy named for testing (the agent only ever works on a copy; the demo calls it BHO_SBX).
3. Note the trial's limits: saving, exporting, Excel linking and any project-size limits may be restricted in a trial.

[confirm in your trial](#)

2.2 The control workbook

1. Create an Excel workbook, e.g. `C:\Bohio\AlNarjis_control.xlsx`, with two sheets: **Inputs** and **Outputs**.
2. On **Inputs**, one row per assumption the agent may change: label in column B, value in column C. Start with the lines in `runner/register_map.csv` (33 lines: unit prices, rents, construction lines, fees, contingency, GFA, delay, SAIBOR, margin, LTC, fund terms). Only include what your model actually has.
3. On **Outputs**, one row per result: levered and unlevered IRR, profit on cost, net profit, total cost, gross revenue, peak debt, equity multiple.
4. Use EstateMaster's Excel link to link each input in the model to its cell on **Inputs**, and each output on **Outputs** to the model's result. [confirm in your trial](#)

2.3 Map the register

1. Fill `runner/register_map.csv`: for each line put the sheet and cell (e.g. `c4, Inputs, C12`) and, for your own reference, where it lives in EstateMaster.
2. Fill `runner/outputs_map.csv` the same way for the outputs.
3. Delete lines your model does not have; add lines it has that are missing (keep ids short and unique).

2.4 Prove the round trip by hand, and time it

1. Change one input in the workbook (e.g. MEP cost +10%). Save.
2. Refresh the link in EstateMaster so the model picks the value up and recalculates. Note exactly what you had to do (automatic, a button, a menu command). [confirm in your trial](#)
3. Check that the Outputs sheet shows the new IRR. Note how long the whole round trip took.
4. Repeat with three or four different inputs, including a timing input (delay) and a finance input (LTC).

Questions for Altus from this phase. Can the live-link be refreshed without a person clicking (by a macro or command)? Is unattended use on a server allowed under the licence? Which inputs can be linked (unit mix, cost lines, timing, funding)? Can report tables be exported to Excel automatically? Does the SQL Server sync cover feasibility data or only accounting and sales CRM? Can a project file be copied and its data cleared for use as a template?

Phase 3: Run the EstateMaster runner

The runner is a small web service on the EstateMaster machine. The agent (or you, for testing) sends it inputs; it writes them to the control workbook, triggers the refresh and returns the outputs. The starter kit is in `runner/em_runner.py`; it has not been run against EstateMaster, and its refresh step must match what you found in 2.4.

1. Copy the `runner` folder to the EstateMaster machine, e.g. `C:\Bohio\runner`.
2. Open PowerShell there and install the dependencies: `py -m pip install fastapi uvicorn xlwings`.
3. Set the variables: `$env:RUNNER_TOKEN="<long random string>"` and `$env:CONTROL_WORKBOOK="C:\Bohio\AlNarjis_control.xlsx"`.
4. Edit `refresh_estatemaster()` in `em_runner.py` to do what refreshed the model in 2.4: recalculate and save, run a macro (`wb.macro("RefreshEstateMaster")()`), or, only as a last resort, drive the EstateMaster screen.
confirm in your trial
5. Start it: `py -m uvicorn em_runner:app --host 127.0.0.1 --port 8765`.
6. Test from the same machine: `curl -H "Authorization: Bearer <token>" http://127.0.0.1:8765/health`, then post a change to `/run` with body `{{"inputs":{{"c4":1430}}}}` and check the returned outputs and seconds.
7. To let Vercel reach it later, expose it through a tunnel rather than opening a port, e.g. Cloudflare Tunnel: `cloudflared tunnel --url http://localhost:8765`. Keep the token secret; the tunnel URL is not a password.

Leave Excel and EstateMaster signed in on that machine; the runner uses the running Excel instance.

Phase 4: Connect Outlook (Microsoft Graph)

The agent reads one mailbox folder, read-only, and never sends or deletes mail. Everything it finds becomes a change request that a person approves.

4.1 Choose the mailbox and folder

1. Use a project mailbox (a shared mailbox such as `al-narjis@kinan.com`) or a folder in an existing mailbox, e.g. a folder called **Al Narjis** at the top level or under Inbox.
2. Route project mail there (forwarding rule, or ask consultants to copy the project address).

4.2 Register the app in Microsoft Entra ID (admin)

1. Go to the Microsoft Entra admin center → **App registrations** → **New registration**. Name: *Bohio agent – Outlook reader*. Supported account types: this organisation only. No redirect URI.
2. Copy the **Application (client) ID** and the **Directory (tenant) ID**.
3. **API permissions** → Add a permission → Microsoft Graph → **Application permissions** → `Mail.Read` → Add. Then **Grant admin consent**.
4. **Certificates & secrets** → New client secret. Copy the value now (it is shown once). Note its expiry date.

4.3 Restrict the app to the project mailbox (strongly recommended)

Mail.Read as an application permission can read every mailbox until you restrict it. In Exchange Online PowerShell, as an Exchange administrator:

```
Connect-ExchangeOnline
# a mail-enabled security group containing only the project mailbox
New-ApplicationAccessPolicy -AppId <client-id> -PolicyScopeGroupId bohio-mailboxes@kinan.com -AccessRight
RestrictAccess -Description "Bohio agent: project mailbox only"
Test-ApplicationAccessPolicy -Identity al-narjis@kinan.com -AppId <client-id> # should say Granted
Test-ApplicationAccessPolicy -Identity someone.else@kinan.com -AppId <client-id> # should say Denied
```

Microsoft now also offers Role Based Access Control for Applications in Exchange Online as the newer way to scope an app to specific mailboxes; your administrator may prefer it. Either way, test that other mailboxes are denied.

4.4 Add the settings to Vercel and test

VARIABLE	VALUE
MS_TENANT_ID	Directory (tenant) ID
MS_CLIENT_ID	Application (client) ID
MS_CLIENT_SECRET	the secret value
OUTLOOK_MAILBOX	e.g. al-narjis@kinan.com
OUTLOOK_FOLDER	folder display name, e.g. Al Narjis (default Inbox)
EXTRACT_MODEL	optional model override for extraction

1. Redeploy. Open `https://<your-site>/api/scan` : it should show `"configured":true` with your mailbox and folder.
2. In the demo, the Outlook card in the header changes to "Live · <mailbox>".
3. Send a test email to the folder, e.g. "Following the chiller redesign our MEP price moves to SAR 1,420 per sqm GFA."
4. Open the Outlook tab and press **Scan now**. The email appears with the extracted change and quote, and a change request appears in Approvals. Approve it: MEP updates and the change memory records who approved what.
5. Emails with no assumption content are listed but ignored.

4.5 Four scans a day

While the workspace is open, it scans at 06:00, 10:00, 14:00 and 18:00 Riyadh time and on request, and runs the assumption review (market, actuals, sales, cost library, zoning) at the same time. For scans with nobody logged in, add a Vercel Cron job calling a scan endpoint at 03:00, 07:00, 11:00 and 15:00 UTC (Riyadh is UTC+3) and a database to hold the proposals (e.g. Vercel Postgres, or SQL Server in production). That storage step is not in the demo yet.

Data protection. Email content is sent to the AI provider for extraction. Confirm with KINAN's compliance team (Saudi PDPL) which provider and region are acceptable, keep the folder limited to project mail, and rotate the client secret before it expires.

Phase 5: Optional pieces

WhatsApp

- The demo simulates WhatsApp. For the real channel: a Meta Business account, a WhatsApp Business Account on the WhatsApp Business Platform (Cloud API), a dedicated phone number, preferably through a Business Solution Provider, approved message templates for scheduled alerts, and a webhook on Vercel that passes messages to the agent.
- Keep an allow-list of numbers mapped to roles (Analyst, Investment Director) and require a one-time code for approvals that change the live model. The webhook is not built yet.

Excel context and Microsoft 365 Copilot

- In the demo you can already attach Excel files (+ Excel) and ask about them.

- For Copilot: a Copilot Studio agent calling the same tools through a custom connector (OpenAPI) or an MCP server, with files read from SharePoint/OneDrive under each user's permissions. Not built yet.

Connecting the agent's calculations to the runner (next development step)

- Add a `RUNNER_URL` and `RUNNER_TOKEN` to Vercel and a small `/api/runner` proxy.
- Route approved changes, "Verify in EstateMaster", extracts and reports to the runner; keep the replica for fast screening only.
- Store every runner job (inputs, outputs, time, who asked) with the change memory.

Go-live checklist

- Access code set; API keys only in Vercel, never in the page.
- Outlook app restricted to the project mailbox and tested (other mailboxes denied).
- Client secret expiry in the calendar.
- Runner machine: Windows updates, Excel and EstateMaster signed in, runner started as a service, token stored securely, tunnel running.
- Altus has confirmed licence terms for unattended use.
- KINAN has confirmed hurdles, stress scenarios, report templates and data residency.
- Diagnostics (Router → Run diagnostics) all pass on the deployed site.

Troubleshooting

SYMPTOM	LIKELY CAUSE AND FIX
Router says "rules only (no AI connected)"	No API key in Vercel, or not redeployed after adding it. Check <code>/api/llm</code> shows the provider as true.
"Access code missing or wrong"	DEMO_PASSWORD is set: enter the code under Router.
<code>/api/scan</code> shows <code>configured:false</code>	One of MS_TENANT_ID, MS_CLIENT_ID, MS_CLIENT_SECRET, OUTLOOK_MAILBOX or an AI key is missing.
"Microsoft sign-in failed"	Wrong tenant or client ID, secret expired or copied as the secret ID instead of the value.
"Graph 403"	Admin consent not granted for Mail.Read, or the access policy does not include this mailbox.
"Folder ... not found"	OUTLOOK_FOLDER must match the display name exactly; top level or directly under Inbox.
Emails read but no changes found	The email does not state a new value, or the register line is not in the model. Check the Outlook tab: the email shows "no assumption".
Runner <code>/run</code> returns old outputs	The refresh step did not run or needs longer: adjust <code>refresh_estatemaster()</code> or <code>REFRESH_WAIT_S</code> .
Runner "not in register_map.csv"	Add the line id to the map, or remove it from the request.
Excel download or attach fails	The page loads the Excel reader from cdnjs.cloudflare.com; allow that host.

Appendix: files in the zip

FILE	PURPOSE
index.html	The whole demo; dummy data and Excel embedded
api/llm.js	AI proxy: Claude and OpenAI with server-side keys, access code, output cap

api/scan.js	Outlook through Microsoft Graph, AI extraction of assumption changes
runner/em_runner.py	EstateMaster runner starter kit (FastAPI + xlwings)
runner/register_map.csv	The 33-line assumption register to map to your workbook
runner/outputs_map.csv	Outputs to map
vercel.json, package.json, .vercelignore	Vercel settings
data/	Scripts that generate the dummy dataset (not deployed)
README.md	Summary of all of the above